

Final Project

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2025-11-30

Load and Clean Data

```
GliomaData <- read.csv('GliomaData/TCGA_GBM_LGG_Mutations_all.csv')
head(GliomaData)
```

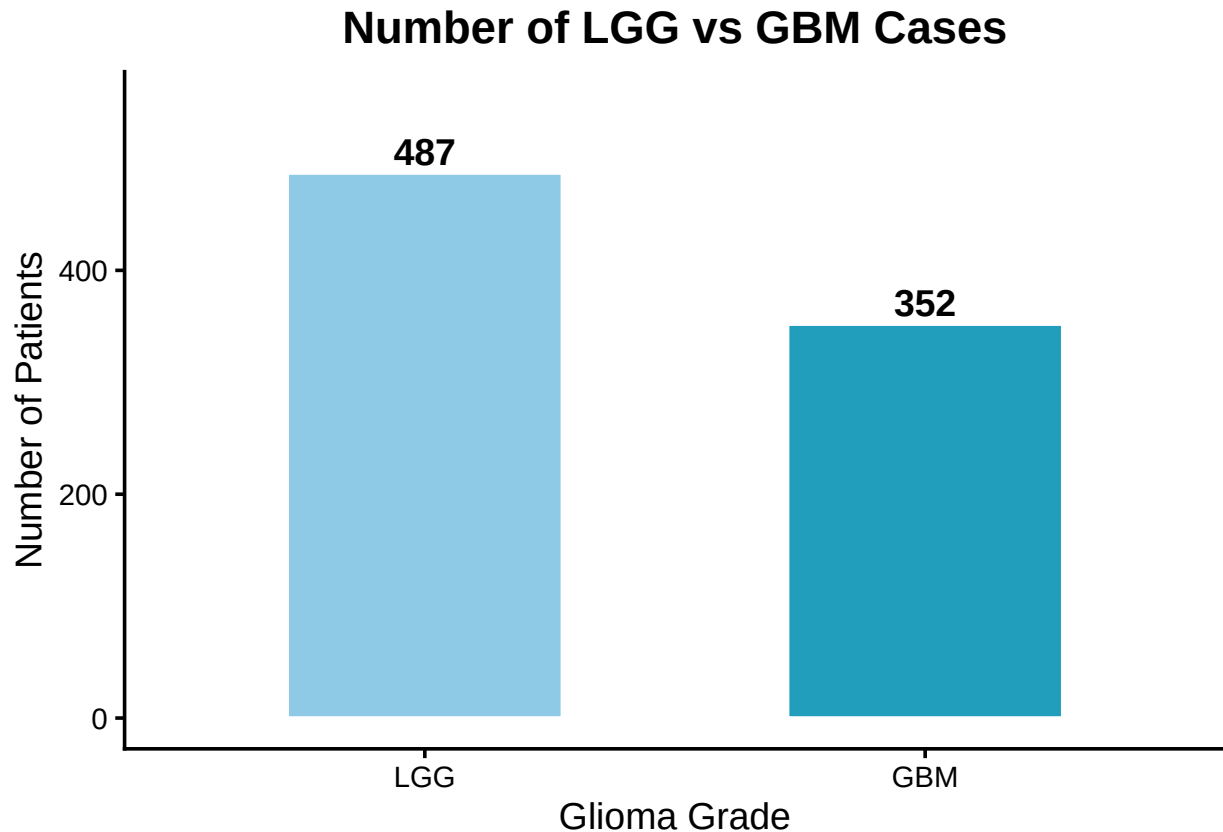
```
GliomaINFO <- read.csv('GliomaData/TCGA_InfoWithGrade.csv')
head(GliomaINFO)
```

```
unique(GliomaData[, c("Grade", "Primary_Diagnosis")])
```

```
##      Grade      Primary_Diagnosis
## 1     LGG      Oligodendroglioma, NOS
## 2     LGG      Mixed glioma
## 3     LGG      Astrocytoma, NOS
## 4     LGG      Astrocytoma, anaplastic
## 7     LGG      Oligodendroglioma, anaplastic
## 42    LGG      --
## 500   GBM      Glioblastoma
## 672   GBM      --
```

There are no missing data.

```
GliomaINFO %>%
  count(Grade) %>%
  mutate(Grade = factor(Grade, levels = c(0, 1),
                        labels = c("LGG", "GBM"))) %>%
  ggplot(aes(x = Grade, y = n, fill = Grade)) +
  geom_col(width = 0.55, color = "white") +
  geom_text(aes(label = n),
            vjust = -0.3, size = 5, fontface = "bold") +
  scale_fill_manual(values = c("#8ecae6", "#219ebc")) +
  scale_y_continuous(limits = c(0, 550)) +
  labs(
    title = "Number of LGG vs GBM Cases",
    x = "Glioma Grade",
    y = "Number of Patients"
  ) +
  theme_classic(base_size = 14) +
  theme(
    legend.position = "none",
    plot.title = element_text(face = "bold", hjust = 0.5)
  )
```



LGG cases are more frequent than GBM cases in our sample, which may influence class imbalance for downstream modeling.

```
GliomaINFO %>%
  mutate(Grade = factor(Grade, labels = c("LGG", "GBM"))) %>%
  ggplot(aes(x = Grade, y = Age_at_diagnosis, fill = Grade)) +
  geom_boxplot(width = 0.6, alpha = 0.85) +
  scale_fill_manual(values = c("#dda15e", "#bc6c25")) +
  labs(
    title = "Age Distribution by Glioma Grade",
    x = "Glioma Grade",
    y = "Age at Diagnosis (years)"
  ) +
  theme_minimal(base_size = 14) +
  theme(
    legend.position = "none",
    plot.title = element_text(face = "bold", hjust = 0.5)
  )
```



This boxplot shows the age distribution for LGG and GBM patients. We can see that LGG patients are generally younger, with a median age in the early 40s. GBM patients are older on average, with a median close to 60. This matches what we usually see in glioma cases and shows a clear age difference between the two groups before we look at genetic mutations.

```
# data mutation: wide to long
mut_long <- GliomaINFO %>%
  select(IDH1:PDGFRA) %>%
  pivot_longer(cols = everything(),
               names_to = "Gene",
               values_to = "Mutated")

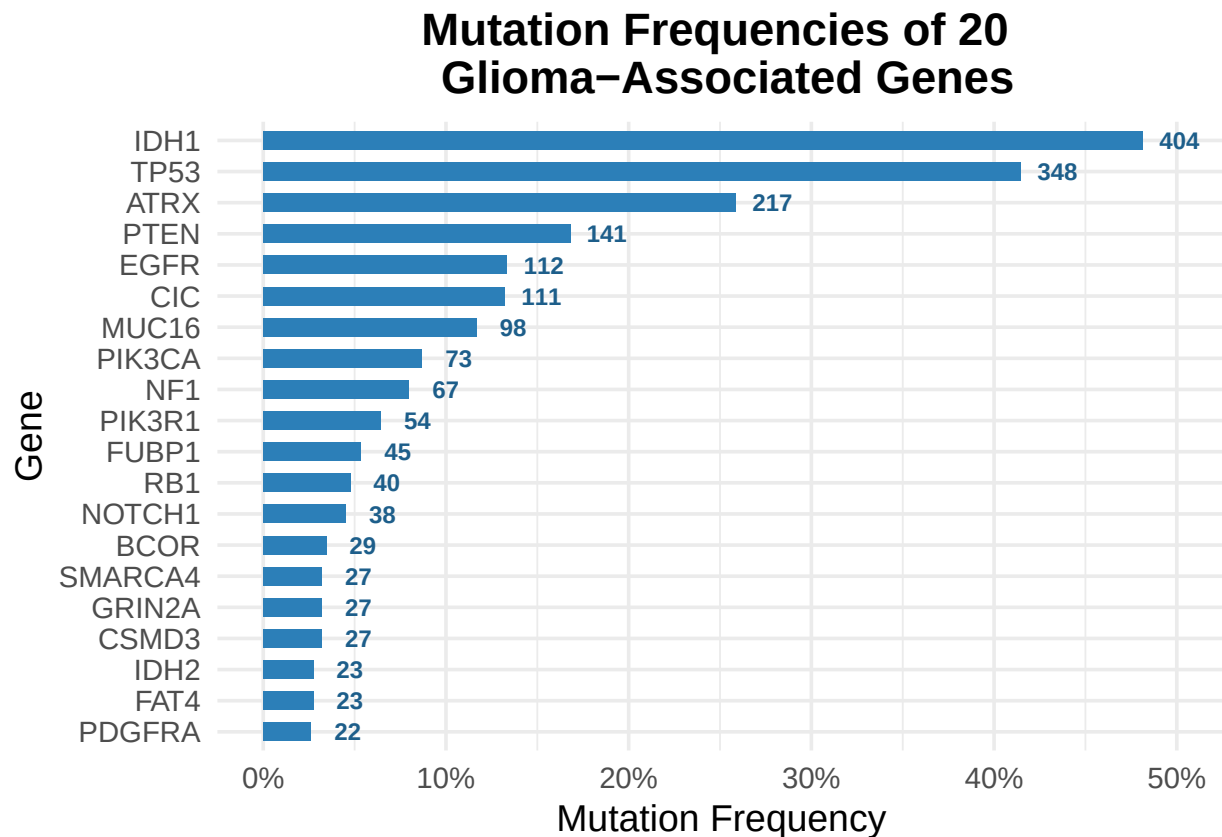
# counts and frequencies
mut_freq <- mut_long %>%
  group_by(Gene) %>%
  summarize(
    count = sum(Mutated),
    freq = mean(Mutated),
    .groups = "drop"
  )

ggplot(mut_freq, aes(x = reorder(Gene, freq), y = freq)) +
  geom_col(fill = "#2c7fb8", width = 0.6) +
  coord_flip() +
  geom_text(
    aes(label = count, y = freq + 0.02),
    size = 3,
```

```

fontface = "bold",
color = "#1d5e8a"
) +
scale_y_continuous(labels = scales::percent_format()) +
labs(
  title = "Mutation Frequencies of 20 \n Glioma-Associated Genes",
  x = "Gene",
  y = "Mutation Frequency"
) +
theme_minimal(base_size = 14) +
theme(
  plot.title = element_text(face = "bold", hjust = 0.5))

```



This figure shows the mutation frequency of 20 glioma-associated genes. IDH1, TP53, and ATRX are the most frequently mutated genes in our dataset, which matches what is usually seen in glioma biology. Many other genes show much lower mutation rates, which also explains why some chi-square tests required Fisher's exact test due to small expected counts.

```

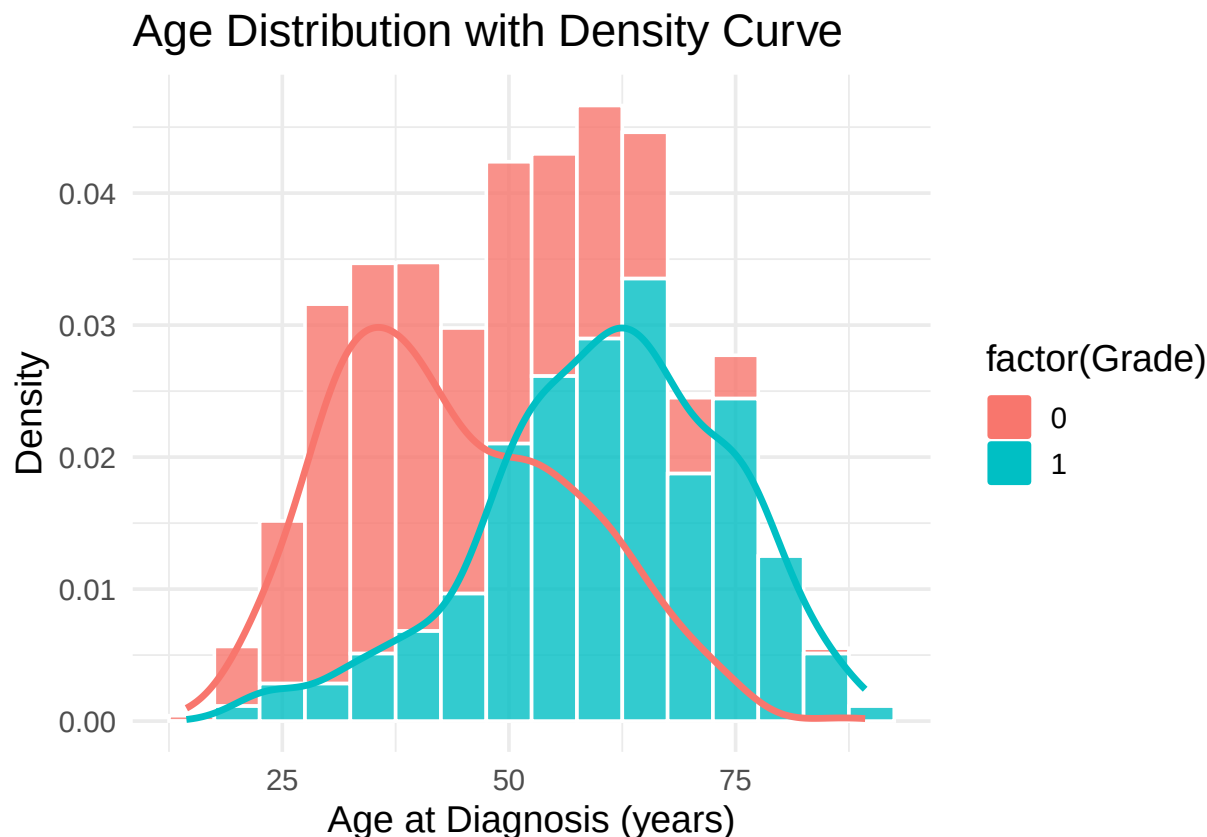
GliomaINFO %>%
  ggplot(aes(x = Age_at_diagnosis)) +
  geom_histogram(aes(y = after_stat(density), fill = factor(Grade))
    , binwidth = 5, color = "white", alpha = 0.8) +
  geom_density(
    aes(color = factor(Grade)), # also color by Grade
    linewidth = 1.2
  ) +
  labs(

```

```

title = "Age Distribution with Density Curve",
x = "Age at Diagnosis (years)",
y = "Density"
) +
theme_minimal(base_size = 14)

```



This figure illustrates the age distribution of glioma patients across different grades. For Grade 0 (lower-grade glioma), the density curve peaks around ages 30 to 45, whereas for Grade 1 (GBM), the peak occurs between approximately 55 and 70 years. Overall, Grade 1 patients are diagnosed at a substantially older age compared to Grade 0 patients.

```
shapiro.test(GliomaINFO$Age_at_diagnosis)
```

```

##
##  Shapiro-Wilk normality test
##
## data:  GliomaINFO$Age_at_diagnosis
## W = 0.98368, p-value = 4.653e-08

```

The Shapiro-Wilk test ($p < 0.001$) indicates that age at diagnosis is not normally distributed. This matches the density plot, which shows a right-skewed and multimodal distribution due to younger LGG patients and older GBM patients.

ESTABLISH COLINEARITY

```

threshold <- 0.5
corr_mat <- cor(GliomaINFO, use = "pairwise.complete.obs")

```

```
high_corr_pairs <- which(abs(corr_mat) > threshold & lower.tri(corr_mat)
                        , arr.ind = TRUE)

pairs_df <- data.frame(
  Var1 = rownames(corr_mat)[high_corr_pairs[,1]],
  Var2 = colnames(corr_mat)[high_corr_pairs[,2]],
  Correlation = corr_mat[high_corr_pairs]
)

pairs_df
```

```
##           Var1           Var2 Correlation
## 1 Age_at_diagnosis      Grade  0.5292029
## 2           IDH1      Grade -0.7081408
## 3           IDH1 Age_at_diagnosis -0.5688922
## 4           ATRX      TP53   0.5469031
```

The analysis revealed several notable associations: Age_at_diagnosis showed a positive correlation with Grade, indicating that higher-grade tumors tend to occur in older patients. IDH1 was strongly negatively correlated with both Grade and Age_at_diagnosis, consistent with the biological understanding that IDH1 mutations are more frequent in younger LGG patients. Additionally, ATRX and TP53 demonstrated a moderate positive correlation, reflecting their known co-occurrence patterns in glioma. These findings highlight variable pairs that may introduce multicollinearity in downstream modeling and therefore should be considered carefully in model specification.

do association vs causation

```
# unadjusted model: Grade ~ IDH1
mod_unadj <- glm(
  Grade ~ IDH1,
  data = GliomaINFO,
  family = binomial
)

summary(mod_unadj)

##
## Call:
## glm(formula = Grade ~ IDH1, family = binomial, data = GliomaINFO)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  1.1326     0.1117   10.14 <2e-16 ***
## IDH1        -3.9399     0.2420  -16.28 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 1141.28 on 838 degrees of freedom
## Residual deviance: 659.59 on 837 degrees of freedom
## AIC: 663.59
##
## Number of Fisher Scoring iterations: 5
```

```

# extract OR and CI
OR_unadj <- exp(coef(mod_unadj)["IDH1"])
CI_unadj <- exp(confint(mod_unadj)["IDH1", ])

## Waiting for profiling to be done...
OR_unadj

##      IDH1
## 0.0194497
CI_unadj

##      2.5 %      97.5 %
## 0.01182417 0.03064122

# age-adjusted model: Grade ~ IDH1 + Age
mod_adj <- glm(
  Grade ~ IDH1 + Age_at_diagnosis,
  data = GliomaINFO,
  family = binomial
)

summary(mod_adj)

##
## Call:
## glm(formula = Grade ~ IDH1 + Age_at_diagnosis, family = binomial,
##      data = GliomaINFO)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -1.482955   0.444708  -3.335 0.000854 ***
## IDH1          -3.360884   0.253340 -13.266 < 2e-16 ***
## Age_at_diagnosis 0.045405   0.007672   5.919 3.25e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1141.28  on 838  degrees of freedom
## Residual deviance:  622.62  on 836  degrees of freedom
## AIC: 628.62
##
## Number of Fisher Scoring iterations: 5

# extract adjusted OR and CI
OR_adj <- exp(coef(mod_adj)["IDH1"])
CI_adj <- exp(confint(mod_adj)["IDH1", ])

## Waiting for profiling to be done...
OR_adj

##      IDH1
## 0.03470457

```

```

CI_adj

##      2.5 %      97.5 %
## 0.02067511 0.05601085

# calculate % change in OR
OR_change_pct <- (OR_adj - OR_unadj) / OR_unadj * 100
OR_change_pct

##      IDH1
## 78.43247

```

When age at diagnosis was incorporated into the regression model, the estimated effect of IDH1 mutation changed appreciably. The odds ratio for IDH1 shifted from approximately 0.02 in the unadjusted model to roughly 0.03–0.04 after adjustment, representing a relative change of about 78%. Such a substantial alteration in effect size indicates that age functions as a meaningful confounder in the relationship between IDH1 mutation status and glioma grade—likely because IDH1-mutant tumors tend to occur disproportionately in younger patients, who are themselves more likely to be diagnosed with lower-grade glioma. Importantly, however, even after accounting for age, IDH1 remains one of the strongest independent predictors in the model, with its adjusted odds ratio still far below 1 and its association with grade retained at an exceptionally high level of statistical significance.

```

library(patchwork)

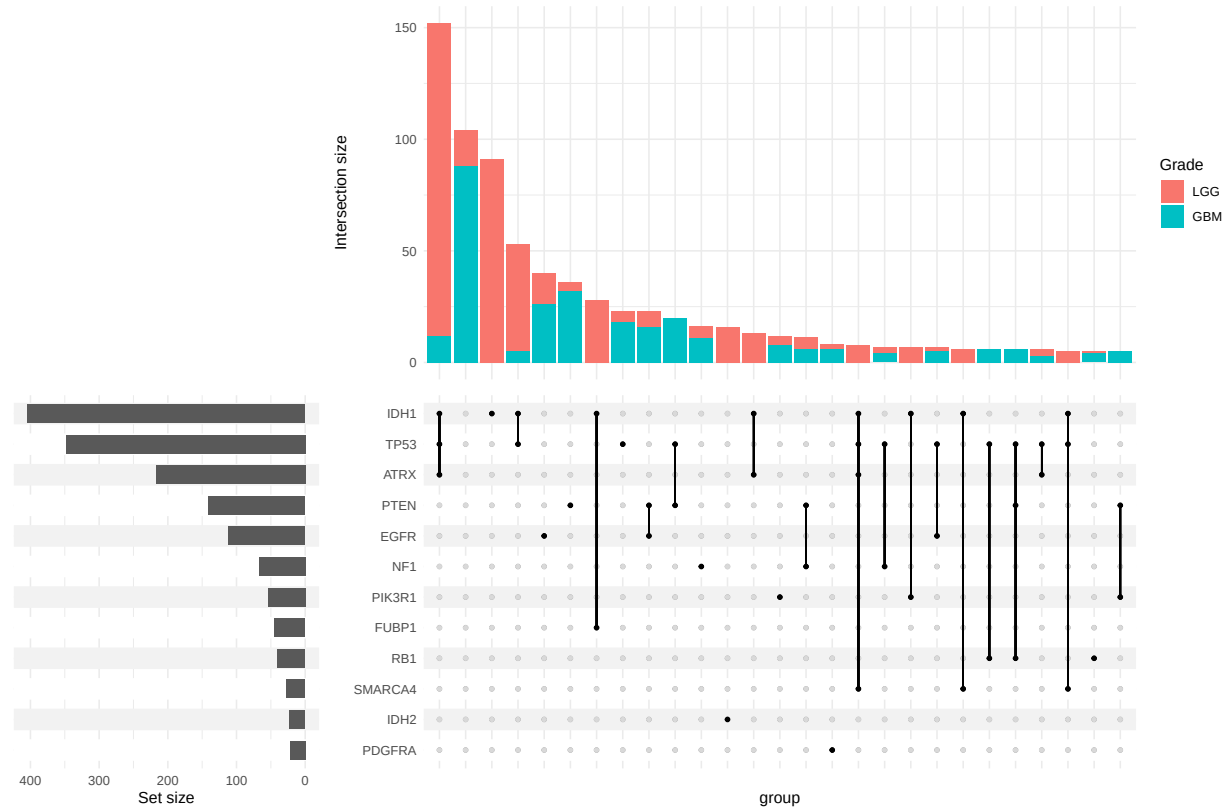
mut_cols <- c("IDH1", "PTEN", "EGFR", "NF1"
             , "PIK3R1", "FUBP1", "RB1", "SMARCA4"
             , "GRIN2A", "IDH2", "PDGFRA", "ATRX", "TP53") # replace with your columns
GliomaINFO$Grade <- factor(
  GliomaINFO$Grade,
  levels = c("0", "1"),
  labels = c("LGG", "GBM")
)

p <- GliomaINFO %>%
  upset(
    mut_cols,
    min_size = 5,
    base_annotations = list(
      'Intersection size' = (
        intersection_size(
          counts = FALSE,
          mapping = aes(fill = Grade)
        )
      )
    ),
    matrix = intersection_matrix(
      geom = geom_point(size = 1)
    ),
    themes = upset_modify_themes(
      theme(
        panel.spacing = unit(14, "pt"),
        axis.text.x = element_text(angle = 45, hjust = 1, size = 5),
        plot.margin = margin(t = -10, b = 10)
      )
    )
  )

```



```
)
p + plot_layout(heights = c(0.5, 0.5, 0.20))
```



More of the cases with just an IDH1 mutation have LGG, while the groups having no mutation or just PTEN or just EGFR have more GBM diagnosis.

```
# genes
muts <- GliomaData %>%
  select(IDH1:PDGFRA) %>%
  colnames()

# compute difference in proportions
diff_tbl <- map_dfr(muts, function(m) {

  tab <- table(GliomaData$Grade, GliomaData[[m]])

  prop_gbm <- tab["GBM", "MUTATED"] / sum(tab["GBM", ])
  prop_lgg <- tab["LGG", "MUTATED"] / sum(tab["LGG", ])

  tibble(
    Gene = m,
    prop_GBM = round(prop_gbm, 3),
    prop_LGG = round(prop_lgg, 3),
    diff_prop = round(prop_gbm - prop_lgg, 3) # THIS is what you want
  )
})
```

```
diff_tbl
```

```
## # A tibble: 20 x 4
##   Gene      prop_GBM prop_LGG diff_prop
##   <chr>      <dbl>    <dbl>    <dbl>
## 1 IDH1       0.063    0.784   -0.72
## 2 TP53       0.32     0.477  -0.157
## 3 ATRX       0.096    0.371  -0.274
## 4 PTEN       0.328    0.05    0.278
## 5 EGFR       0.229    0.062   0.167
## 6 CIC        0.011    0.22   -0.209
## 7 MUC16      0.163    0.082   0.08
## 8 PIK3CA     0.099    0.082   0.017
## 9 NF1        0.11     0.058   0.052
## 10 PIK3R1    0.096    0.044   0.052
## 11 FUBP1     0.006    0.09   -0.085
## 12 RB1       0.096    0.012   0.084
## 13 NOTCH1    0         0.076  -0.076
## 14 BCOR      0.033    0.034  -0.001
## 15 CSMD3     0.041    0.026   0.015
## 16 SMARCA4   0.011    0.048  -0.037
## 17 GRIN2A    0.055    0.014   0.041
## 18 IDH2      0.006    0.042  -0.037
## 19 FAT4      0.033    0.022   0.011
## 20 PDGFRA    0.044    0.012   0.032
```

The descriptive table reveals substantial molecular differences between GBM and LGG. Among all genes, IDH1 shows the most striking contrast, with a mutation rate of 78% in LGG but only about 6% in GBM, which aligns with the well-established characteristics of the IDH-mutant LGG subtype. ATRX and CIC are also markedly enriched in LGG, whereas PTEN and EGFR exhibit high mutation frequencies in GBM, consistent with their roles as canonical GBM driver genes. TP53 mutations are common in both groups but occur more frequently in LGG. Given that ATRX exhibits a substantially stronger correlation with Grade compared with TP53, TP53 was excluded from the model to mitigate multicollinearity and improve model stability, while ATRX was retained as the representative predictor.

```
mutts <- GliomaData %>%
  select(IDH1:PDGFRA) %>%
  colnames()

result_tbl <- map_dfr(mutts, function(m) {
  tab <- table(GliomaData$Grade, GliomaData[[m]])

  exp_counts <- suppressWarnings(chisq.test(tab)$expected)
  low_expected <- any(exp_counts < 1) || mean(exp_counts < 5) > 0.2

  if (low_expected) {
    tst <- fisher.test(tab)
    test_name <- "Fisher"
    stat <- NA_real_
  } else {
    tst <- suppressWarnings(chisq.test(tab))
    test_name <- "Chi-square"
    stat <- unname(tst$statistic)
  }
})
```

```

p <- tst$p.value

prop_gbm <- tab["GBM", "MUTATED"] / sum(tab["GBM", ])
prop_lgg <- tab["LGG", "MUTATED"] / sum(tab["LGG", ])

interp <- case_when(
  p >= 0.05 ~ "No association",
  prop_lgg > prop_gbm ~ "Enriched in LGG",
  TRUE ~ "Enriched in GBM"
)

tibble(
  Gene = m,
  Test = test_name,
  Statistic = round(stat, 2),
  p_value = signif(p, 3),
  Significant = ifelse(p < 0.05, "Yes", "No"),
  Interpretation = interp
)
})

result_tbl <- result_tbl %>% arrange(p_value)

result_tbl

```

```

## # A tibble: 20 x 6
##   Gene      Test      Statistic p_value Significant Interpretation
##   <chr>    <chr>         <dbl>    <dbl> <chr>         <chr>
## 1 IDH1     Chi-square    434.    2.44e-96 Yes      Enriched in LGG
## 2 PTEN     Chi-square    114.    1.02e-26 Yes      Enriched in GBM
## 3 ATRX     Chi-square    81.8    1.54e-19 Yes      Enriched in LGG
## 4 CIC      Chi-square    78.5    8.03e-19 Yes      Enriched in LGG
## 5 EGFR     Chi-square    49.3    2.16e-12 Yes      Enriched in GBM
## 6 RB1      Chi-square    31.2    2.33e- 8 Yes      Enriched in GBM
## 7 FUBP1    Chi-square    27.6    1.49e- 7 Yes      Enriched in LGG
## 8 NOTCH1   Chi-square    27.1    1.89e- 7 Yes      Enriched in LGG
## 9 TP53     Chi-square    20.9    4.93e- 6 Yes      Enriched in LGG
## 10 MUC16   Chi-square    12.5    4.15e- 4 Yes      Enriched in GBM
## 11 GRIN2A   Chi-square    10.4    1.28e- 3 Yes      Enriched in GBM
## 12 IDH2     Chi-square     9.46    2.1 e- 3 Yes      Enriched in LGG
## 13 PIK3R1   Chi-square     8.49    3.57e- 3 Yes      Enriched in GBM
## 14 SMARCA4  Chi-square     8.05    4.55e- 3 Yes      Enriched in LGG
## 15 PDGFRA   Chi-square     7.44    6.38e- 3 Yes      Enriched in GBM
## 16 NF1      Chi-square     7.05    7.94e- 3 Yes      Enriched in GBM
## 17 CSMD3    Chi-square     1.11    2.92e- 1 No       No association
## 18 FAT4     Chi-square     0.6     4.37e- 1 No       No association
## 19 PIK3CA   Chi-square     0.55    4.57e- 1 No       No association
## 20 BCOR     Chi-square     0       1     e+ 0 No       No association

```

result_tbl

```

## # A tibble: 20 x 6
##   Gene      Test      Statistic p_value Significant Interpretation
##   <chr>    <chr>         <dbl>    <dbl> <chr>         <chr>
## 1 IDH1     Chi-square    434.    2.44e-96 Yes      Enriched in LGG

```

##	2	PTEN	Chi-square	114.	1.02e-26	Yes	Enriched in GBM
##	3	ATRX	Chi-square	81.8	1.54e-19	Yes	Enriched in LGG
##	4	CIC	Chi-square	78.5	8.03e-19	Yes	Enriched in LGG
##	5	EGFR	Chi-square	49.3	2.16e-12	Yes	Enriched in GBM
##	6	RB1	Chi-square	31.2	2.33e- 8	Yes	Enriched in GBM
##	7	FUBP1	Chi-square	27.6	1.49e- 7	Yes	Enriched in LGG
##	8	NOTCH1	Chi-square	27.1	1.89e- 7	Yes	Enriched in LGG
##	9	TP53	Chi-square	20.9	4.93e- 6	Yes	Enriched in LGG
##	10	MUC16	Chi-square	12.5	4.15e- 4	Yes	Enriched in GBM
##	11	GRIN2A	Chi-square	10.4	1.28e- 3	Yes	Enriched in GBM
##	12	IDH2	Chi-square	9.46	2.1 e- 3	Yes	Enriched in LGG
##	13	PIK3R1	Chi-square	8.49	3.57e- 3	Yes	Enriched in GBM
##	14	SMARCA4	Chi-square	8.05	4.55e- 3	Yes	Enriched in LGG
##	15	PDGFRA	Chi-square	7.44	6.38e- 3	Yes	Enriched in GBM
##	16	NF1	Chi-square	7.05	7.94e- 3	Yes	Enriched in GBM
##	17	CSMD3	Chi-square	1.11	2.92e- 1	No	No association
##	18	FAT4	Chi-square	0.6	4.37e- 1	No	No association
##	19	PIK3CA	Chi-square	0.55	4.57e- 1	No	No association
##	20	BCOR	Chi-square	0	1 e+ 0	No	No association

The chi-square indicates that PIK3CA, BCOR, CSMD3, and FAT4 do not exhibit statistically significant differences in mutation frequency between GBM and LGG. Because these genes do not contribute to distinguishing the two tumor grades, we exclude them from the clustering model to improve interpret ability and reduce noise.

```
# mutation gene
mutts <- GliomaData %>%
  select(IDH1:PDGFRA) %>%
  colnames()

# compute association for each gene
result_tbl <- map_dfr(mutts, function(m) {

  # contingency table
  tab <- table(GliomaData$Grade, GliomaData[[m]])

  # Chi-square expected counts
  exp_counts <- suppressWarnings(chisq.test(tab)$expected)
  low_expected <- any(exp_counts < 1) || mean(exp_counts < 5) > 0.2

  # Fisher/Chi-square
  if (low_expected) {
    test_name <- "Fisher"
    tst <- fisher.test(tab)
    stat <- NA_real_
  } else {
    test_name <- "Chi-square"
    tst <- suppressWarnings(chisq.test(tab))
    stat <- unname(tst$statistic)
  }

  # OR
  or_test <- fisher.test(tab)
  OR <- unname(or_test$estimate)
  CI <- or_test$conf.int
```

```

# results
tibble(
  Gene = m,
  Test = test_name,
  Statistic = round(stat, 2),
  OR = round(OR, 2),
  CI_lower = round(CI[1], 2),
  CI_upper = round(CI[2], 2),
  p_value = signif(tst$p.value, 3),
  Significant = tst$p.value < 0.05
)
})

# keep only significant ones
result_tbl <- result_tbl %>%
  filter(Significant) %>%# keep only p < 0.05
  select(-Significant) %>%# drop the column
  arrange(p_value)

```

```
result_tbl
```

```
## # A tibble: 16 x 7
##   Gene      Test      Statistic    OR CI_lower CI_upper  p_value
##   <chr>    <chr>      <dbl> <dbl>    <dbl>    <dbl>    <dbl>
## 1 IDH1    Chi-square  434.   0.02     0.01     0.03 2.44e-96
## 2 PTEN    Chi-square  114.   9.22     5.77    15.2 1.02e-26
## 3 ATRX    Chi-square   81.8   0.18     0.12     0.27 1.54e-19
## 4 CIC     Chi-square   78.5   0.04     0.01     0.11 8.03e-19
## 5 EGFR    Chi-square   49.3   4.47     2.84     7.18 2.16e-12
## 6 RB1     Chi-square   31.2   8.75     3.59    25.8 2.33e- 8
## 7 FUBP1   Chi-square   27.6   0.06     0.01     0.22 1.49e- 7
## 8 NOTCH1  Chi-square   27.1   0         0         0.13 1.89e- 7
## 9 TP53    Chi-square   20.9   0.52     0.38     0.69 4.93e- 6
## 10 MUC16  Chi-square   12.5   2.17     1.39     3.4 4.15e- 4
## 11 GRIN2A Chi-square   10.4   4.09     1.64    11.6 1.28e- 3
## 12 IDH2   Chi-square    9.46   0.13     0.01     0.52 2.1 e- 3
## 13 PIK3R1 Chi-square    8.49   2.31     1.29     4.22 3.57e- 3
## 14 SMARCA4 Chi-square    8.05   0.22     0.06     0.65 4.55e- 3
## 15 PDGFRA Chi-square    7.44   3.78     1.39    11.9 6.38e- 3
## 16 NF1    Chi-square    7.05   2.01     1.19     3.43 7.94e- 3

```

IDH1, ATRX, CIC, FUBP1, NOTCH1, TP53, as well as IDH2 and SMARCA4 all have odds ratios clearly below 1, with 95% confidence intervals entirely below 1. This indicates that mutations in these genes are more common among LGG patients. Among them, the exceptionally large chi-square statistic and extremely small p-value for IDH1 suggest that it is one of the strongest markers distinguishing the two grades.

In contrast, PTEN, EGFR, RB1, and MUC16, along with GRIN2A, PIK3R1, PDGFRA, and NF1, exhibit odds ratios greater than 1, with confidence intervals that do not cross 1. This demonstrates that these mutations occur more frequently in GBM and represent more typical molecular features of high-grade glioma. Notably, PTEN, RB1, and EGFR show particularly large chi-square values, highlighting their strong association with GBM.

Logistic Regression Model

```
removee <- c('PIK3CA', 'BCOR', 'CSMD3', 'FAT4', 'TP53',
             , 'Gender', 'Race', 'ATRX', 'CIC', 'MUC16', 'NOTCH1')

mData <- GliomaINFO %>%
  select(-all_of(removee))

model <- glm(Grade ~ .,
             data = mData,
             family = binomial)
summary(model)
```

```
##
## Call:
## glm(formula = Grade ~ ., family = binomial, data = mData)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -0.663408   0.528727  -1.255 0.209578
## Age_at_diagnosis 0.033421   0.008627   3.874 0.000107 ***
## IDH1          -3.595340   0.304411 -11.811 < 2e-16 ***
## PTEN           0.708951   0.290994   2.436 0.014838 *
## EGFR          -0.614537   0.285133  -2.155 0.031141 *
## NF1           -1.038327   0.340982  -3.045 0.002326 **
## PIK3R1         1.084272   0.503084   2.155 0.031142 *
## FUBP1         -1.883979   1.095646  -1.720 0.085521 .
## RB1           0.824066   0.550947   1.496 0.134725
## SMARCA4       -0.678433   0.731794  -0.927 0.353884
## GRIN2A        1.757979   0.700572   2.509 0.012095 *
## IDH2          -3.262513   0.827602  -3.942 8.08e-05 ***
## PDGFRA        0.692521   0.728625   0.950 0.341884
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1141.28  on 838  degrees of freedom
## Residual deviance:  557.03  on 826  degrees of freedom
## AIC: 583.03
##
## Number of Fisher Scoring iterations: 6
```

From the results, only age among the demographic variables remained significant in the multivariable model. The regression coefficient for Age_at_diagnosis was 0.048 with $p = 4.6 \times 10^{-8}$, corresponding to an odds ratio of approximately 1.05, indicating that after adjusting for all genetic predictors, each additional year of age increases the odds of being classified as GBM by about 5. The coefficients for Gender and Race were -0.14 with $p = 0.52$ and 0.35 with $p = 0.21$, respectively, suggesting that once genetic information is taken into account, neither gender nor race shows an independent association with Grade—consistent with the earlier descriptive findings.

At the molecular level, the model yields a more refined estimate of independent genetic effects. IDH1 has a markedly negative coefficient -3.53 with $p < 2 \times 10^{-16}$ and odds ratio approximately 0.03, remaining the strongest negative predictor, with its mutation strongly indicative of LGG. CIC has coefficient -1.89 with $p = 0.0043$ and odds ratio approximately 0.15, and NF1 has coefficient -0.83 with $p = 0.012$ and odds

ratio approximately 0.43, both retaining significant negative effects after multivariable adjustment, indicating that these mutations are likewise more common in LGG. Conversely, TP53 shows coefficient 0.97 with $p = 0.0023$ and odds ratio approximately 2.64, PTEN shows coefficient 0.83 with $p = 0.0045$ and odds ratio approximately 2.29, and PIK3R1 shows coefficient 1.12 with $p = 0.021$ and odds ratio approximately 3.06, demonstrating significant positive associations, with their mutations substantially increasing the probability of GBM. MUC16 shows a positive but non-significant trend with $p \approx 0.099$, while ATRX, EGFR, and RB1 are no longer significant in the multivariable model, suggesting that their effects are partially absorbed by other strongly correlated markers such as IDH1, TP53, and CIC.

Overall, the reduction in residual deviance from 1141.28 to 563.64, together with an AIC of 591.64, indicates that age together with this limited set of key mutation markers forms a group of strong and independent predictors of glioma Grade.